

Simulating Hydrocarbon Dynamics With Consistent Force Field Method

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In this paper, we reproduce consistent force field simulations for hydrocarbon molecules as derived by Shneior Lifson and Arieh Warshel in their 1968 paper[1]. Warshel won the 2013 Nobel Prize in Chemistry “for the development of multiscale models for complex chemical systems.” Our simulation was constructed using modern programming techniques and languages such as JAX, a novel Python-based framework. We find similar results as Lifson and Warshel upon simulating ethane, propane, benzene, and isobutane. Investigating these molecules’ energies and bond length dynamics over the first few picoseconds, we gain insight about large-scale and small-scale structural dynamics in hydrocarbons.

Physical simulations are crucial to not only developing new experiments but also testing our understanding of the world. We can compare the results of simulations with reality to revise our understanding, run new experiments, create new simulations, and conjecture about new phenomena. Unlike experimental data, in simulations, we have access to many different quantities that would be extremely difficult or even impossible to measure in real life, such as velocities, positions, energies, etc. In addition, they give a more holistic understanding of what to look for in an experiment.

Molecular simulations are just one of the many types of physical simulations. For molecular simulations, we are interested in operating on the 10^{-10}m scale (angstroms). The forces governing our sample are from quantum mechanics which generally interact over small distances. Large-scale forces such as gravity are negligible. Outside of physics, molecular simulation can be used in drug synthesis.

We based our molecular simulation on the consistent force field (CFF) method. This technique was developed by Shneior Lifson and Arieh Warshel[1], the latter of whom won the Nobel Prize in Chemistry in 2013 “for the development of multiscale models for complex chemical systems.” In this paper, we simulate four hydrocarbons of various sizes and give measurements regarding their geometry.

Consistent force field is a classical simulation of a quantum system that estimates forces between atoms. It operates by calculating the potential of the quantum system and then differentiating with respect to position to determine the force experienced by each object, and then updating positions, velocities, and accelerations per time-step as necessary.

Generally, potential energy of molecules can be written as $V = V_{\text{bonded}} + V_{\text{nonbonded}}$ where $V_{\text{bonded}} = V_{\text{length}} + V_{\text{angle}} + V_{\text{dihedral}}$ refers to potential energy contributed by bonded interactions and $V_{\text{nonbonded}} = V_{\text{electrostatic}} + V_{\text{van der Waals}}$ refers to potential energy contributed by nonbonded interactions.[1] The consistent force field in this paper approximates the bond length and bond angle energy contributions only up to the 2nd order.

To calculate the force field needed for our simulation, we begin by calculating the Lagrangian and solve for its equation of motion. The Lagrangian is kinetic energy minus potential energy:

$$L = T - V,$$

and the equation of motion can be derived by solving

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = \frac{\partial L}{\partial x}.$$

For further discussion about Lagrangians, equations of motions, and general examples, see Appendix A.

First we solve for the potential energy, V , which is only a function of the position matrix, here denoted as $\mathbf{X}$ of atoms. Since we do not yet know the explicit form of this function, we will omit its specific details. Below we have approximated V as a 2nd order Taylor polynomial centered around the position matrix $\mathbf{X}_0$:

$$V(\mathbf{X}) \approx V(\mathbf{X}_0) + \sum_{\alpha} \left(\frac{\partial V}{\partial x_{\alpha}} \right) \delta x_{\alpha} + \frac{1}{2} \sum_{\alpha, \beta} \left(\frac{\partial^2 V}{\partial x_{\alpha} \partial x_{\beta}} \right) \delta x_{\alpha} \delta x_{\beta},$$

where δx_{α} and δx_{β} are equal to $x_{\alpha} - (x_0)_{\alpha}$ and $x_{\beta} - (x_0)_{\beta}$ respectively. For ease, we define $\mathbf{X}_0$ as a position matrix of stable equilibrium. In other words, $\mathbf{X}_0$ represents the

molecule at its ground state. Thus,

$$\begin{aligned} \frac{\partial V}{\partial x_a} &= 0 \\ \Rightarrow V(\mathbf{X}) &\approx V(\mathbf{X}_0) + \frac{1}{2} \sum_{\alpha, \beta} \left(\frac{\partial^2 V}{\partial x_\alpha \partial x_\beta} \right) \delta x_\alpha \delta x_\beta \\ &= V(\mathbf{X}_0) + \frac{1}{2} \delta \mathbf{X}' V''(\mathbf{X}_0) \delta \mathbf{X}, \end{aligned} \quad (1)$$

where $\delta \mathbf{X}' V''(\mathbf{X}_0) \delta \mathbf{X}$ expresses $\sum_{\alpha, \beta} \left(\frac{\partial^2 V}{\partial x_\alpha \partial x_\beta} \right) \delta x_\alpha \delta x_\beta$ in matrix form. Similarly, T , the kinetic energy, is a function of the velocity matrix:

$$T(\dot{\mathbf{X}}) = \frac{1}{2} \dot{\mathbf{X}}' \mathbf{M} \dot{\mathbf{X}}. \quad (2)$$

To demonstrate this is correct, we can expand as follows:

$$\begin{aligned} T(\dot{\mathbf{X}}) &= \frac{1}{2} \dot{\mathbf{X}}' \mathbf{M} \dot{\mathbf{X}} = \frac{1}{2} \begin{bmatrix} \dot{x}_1 & \dot{x}_2 & \dot{x}_3 & \cdots & \dot{x}_n \end{bmatrix} \begin{bmatrix} m_1 & 0 & 0 & \cdots & 0 \\ 0 & m_2 & 0 & \cdots & 0 \\ 0 & 0 & m_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & m_n \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} m_1 \dot{x}_1 & m_2 \dot{x}_2 & m_3 \dot{x}_3 & \cdots & m_n \dot{x}_n \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} \\ &= \frac{1}{2} \left[m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2 + m_3 \dot{x}_3^2 + \cdots + m_n \dot{x}_n^2 \right]. \end{aligned}$$

We substitute the above values into the Lagrangian:

$$\begin{aligned} L &= T - V \\ &\approx \frac{1}{2} \dot{\mathbf{X}}' \mathbf{M} \dot{\mathbf{X}} - V(\mathbf{X}_0) + \frac{1}{2} \delta \mathbf{X}' V''(\mathbf{X}_0) \delta \mathbf{X}, \end{aligned}$$

and then solve for the equation of motion:

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{X}}} \right) &= \frac{\partial L}{\partial \mathbf{X}} \\ \frac{d}{dt} \left(\dot{\mathbf{X}}'^{\circ \frac{1}{2}} \mathbf{M} \dot{\mathbf{X}}^{\circ \frac{1}{2}} \right) &= \delta \mathbf{X}'^{\circ \frac{1}{2}} V''(\mathbf{X}_0) \delta \mathbf{X}^{\circ \frac{1}{2}} \\ \ddot{\mathbf{X}}'^{\circ \frac{1}{2}} \mathbf{M} \dot{\mathbf{X}}^{\circ \frac{1}{2}} &= \delta \mathbf{X}'^{\circ \frac{1}{2}} V''(\mathbf{X}_0) \delta \mathbf{X}^{\circ \frac{1}{2}}, \end{aligned} \quad (3)$$

where $\mathbf{A}^{\circ \frac{1}{2}}$ represents the Hadamard square-root of $\mathbf{A}$.

Given this expression of the force, we calculate the new acceleration $\ddot{x}_n$ and subsequently the change in velocity and the change in position over a Δt . The new position is

$$x_n = x_p + \dot{x}_p \Delta t + \frac{\ddot{x}_n (\Delta t)^2}{2},$$

where x_p and $\dot{x}_p$ are the previous position and velocity respectively. Similarly, the new velocity is

$$\dot{x}_n = \dot{x}_p + \ddot{x}_n \Delta t.$$

With the positions, velocities, and accelerations appropriately updated, we continue to the next time-step where the same process repeats.

Now that we have derived the equations of motions and know how to update the geometry, we must now find the functional form for V . As noted above, we are only going to the 2nd order in the Taylor expansion, which results in a harmonic oscillator. Thus, we must treat each kind of potential, for example the potential over a carbon-carbon bond and a carbon-hydrogen bond, as a harmonic function:

$$\begin{aligned} V(\mathbf{X}) &= \frac{1}{2} \sum_i K_a (a_i - a_0)^2 + \frac{1}{2} \sum_i K_b (b_i - b_0)^2 + \\ &+ \frac{1}{2} \sum_i K_\alpha (\alpha_i - \alpha_0)^2 + \frac{1}{2} \sum_i K_\beta (\beta_i - \beta_0)^2 \\ &+ \frac{1}{2} \sum_i K_\gamma (\gamma_i - \gamma_0)^2 \\ &+ \frac{1}{2} \sum_i K_\phi (1 + \cos(3\phi_i)) \\ &+ V_{\text{nonbonded}}(\mathbf{X}). \end{aligned} \quad (4)$$

Variables are defined in Table I. Figure 1 is a visual rep-

$\mathbf{X}$	Matrix of position vectors of all atoms ($\text{\AA}$)
K_a	Constant for energy contributed by CC bonds ($\text{kcal}/\text{\AA}^2/\text{mol}$)
a_i	Lengths of CC bonds ($\text{\AA}$)
a_0	Default length of CC bonds ($\text{\AA}$)
K_b	Constant for energy contributed by CH bonds ($\text{kcal}/\text{\AA}^2/\text{mol}$)
b_i	Lengths of CH bonds ($\text{\AA}$)
b_0	Default length of CH bonds ($\text{\AA}$)
K_α	Constant for energy contributed by CCC bond angles ($\text{kcal}/\text{rad}^2/\text{mol}$)
α_i	Angles of CCC bonds (rad)
α_0	Default angle of CCC bonds (rad)
K_β	Constant for energy contributed by HCH bond angles ($\text{kcal}/\text{rad}^2/\text{mol}$)
β_i	Angles of HCH bonds (rad)
β_0	Default angle of HCH bonds (rad)
K_γ	Constant for energy contributed by CCH bond angles ($\text{kcal}/\text{rad}^2/\text{mol}$)
γ_i	Angles of CCH bonds (rad)
γ_0	Default angle of CCH bonds (rad)
K_ϕ	Constant for energy contributed by CC bond torsional angles (kcal/mol)
ϕ_i	Torsional angles of CC bonds (rad)
$V_{\text{nonbonded}}$	Nonbonded interaction potential function

TABLE I: Variable definitions.

resentation of what properties are used in equation 4 to calculate the potential energy. The energy contributed

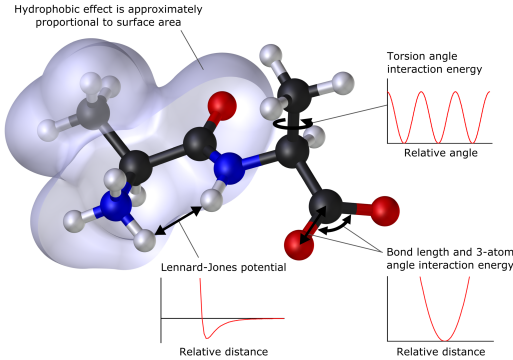


FIG. 1: Illustrative diagram of variables in the potential function, where we ignore the hydrophobic effect.[2]

by bond-lengths is a harmonic, spring-like function:

$$\frac{1}{2} \sum_i K_a (a_i - a_0)^2 \text{ and } \frac{1}{2} \sum_i K_b (b_i - b_0)^2.$$

Energy due to bond angles is also harmonic:

$$\frac{1}{2} \sum_i K_\alpha (\alpha_i - \alpha_0)^2, \frac{1}{2} \sum_i K_\beta (\beta_i - \beta_0)^2, \\ \text{and } \frac{1}{2} \sum_i K_\gamma (\gamma_i - \gamma_0)^2.$$

Torsional angles are treated differently:

$$\frac{1}{2} \sum_i K_\phi (1 + \cos(3\phi_i)).$$

The Lennard-Jones potential shown in the diagram is a part of $V_{\text{nonbonded}}$, as well as electrostatic potential which is not shown. Unfortunately, given the time limits of this project, we were unable to include $V_{\text{nonbonded}}$ entirely in our simulation. However, this the results seem reasonable for near-ground state configurations as will be discussed later.

To produce accurate results, the values of coefficients and constants must be adapted for each system. This requires data and experimentation. Due to time constraints, we chose to use the same values that Lifson and Warshel[1] finalized in their simulation. To arrive at their values, Lifson and Warshel compared the frequencies and strain energies of their simulation and experiments, fine-tuning their coefficients as necessary until an adequate model was established. These coefficients are shown in Table II.

In addition to those constants, it was also necessary to pick initial position matrices for each of the simulated molecules. We chose near-ground state configurations because our simulation is based around a second order Taylor expansion based around the ground state. These

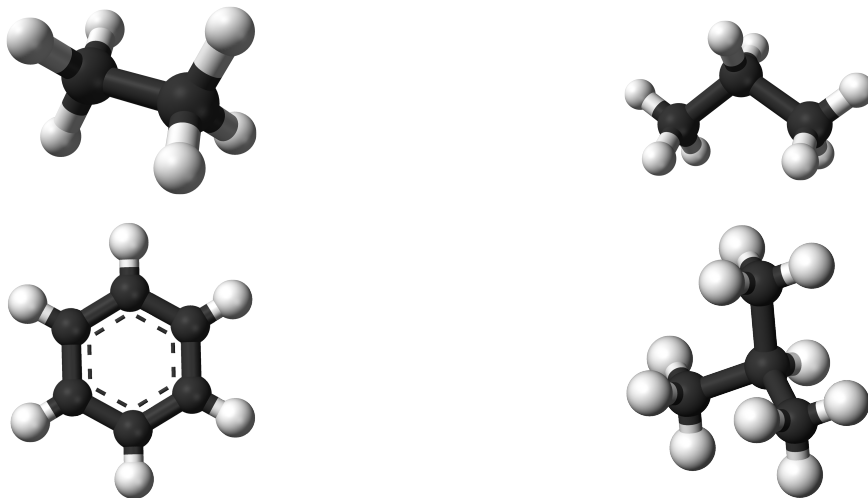


FIG. 2: Diagram of ethane C_2H_6 (top left)[3], diagram of propane C_3H_8 (top right)[4], diagram of benzene C_6H_6 (bottom left)[5], and isobutane $\text{HC}(\text{CH}_3)_3$ (bottom right)[6].

ground state position matrices are listed in Appendix B.

The position matrices we used in our simulation were slightly offset from the ground state geometries using the randomization constant, r . Each atom's initial position is offset by a random vector v of magnitude $0 \leq |v| \leq r$ (if $r = 0$, there is no randomization). We added this randomization constant to simulate a hot molecule. We chose to test the following molecules: ethane C_2H_6 , propane C_3H_8 , benzene C_6H_6 , and isobutane $\text{HC}(\text{CH}_3)_3$ (Figure 2).

For an example of how V is calculated for an ethane molecule at its ground state, examine Appendix C.

Python is advantageous when writing physical simulations in most regards compared to other programming languages due to its many libraries and ease of coding. However, because Python is a scripting language, its most significant drawback is speed. Creating physical simulations with Python is possible and made more accessible using specialized software. Here, we used JAX (a novel Python-based framework) to make this process significantly faster and more accurate. Notably, JAX is an auto-differentiating programming language, meaning it can take the analytical derivative of any given function made with JAX. Furthermore, JAX compiles the code so that it can be run on either the GPU or CPU, some times offering 100 $\times$ speedups. Refer to Appendix D for further details about JAX.

Mass of carbon atom	12.0107 amu
Mass of hydrogen atom	1.00784 amu
Avogadro constant	6.0221409×10^{23}
Δt	1×10^{-18} s
K_a	573.8 kcal/ $\text{\AA}^2/\text{mol}$
a_0	1.455 $\text{\AA}$
K_b	222.0 kcal/ $\text{\AA}^2/\text{mol}$
b_0	1.099 $\text{\AA}$
K_α	53.58 kcal/ rad^2/mol
α_0	1.937 rad
K_β	76.28 kcal/ rad^2/mol
β_0	1.911 rad
K_γ	44.0 kcal/ rad^2/mol
γ_0	1.911 rad
K_ϕ	2.836 kcal/mol

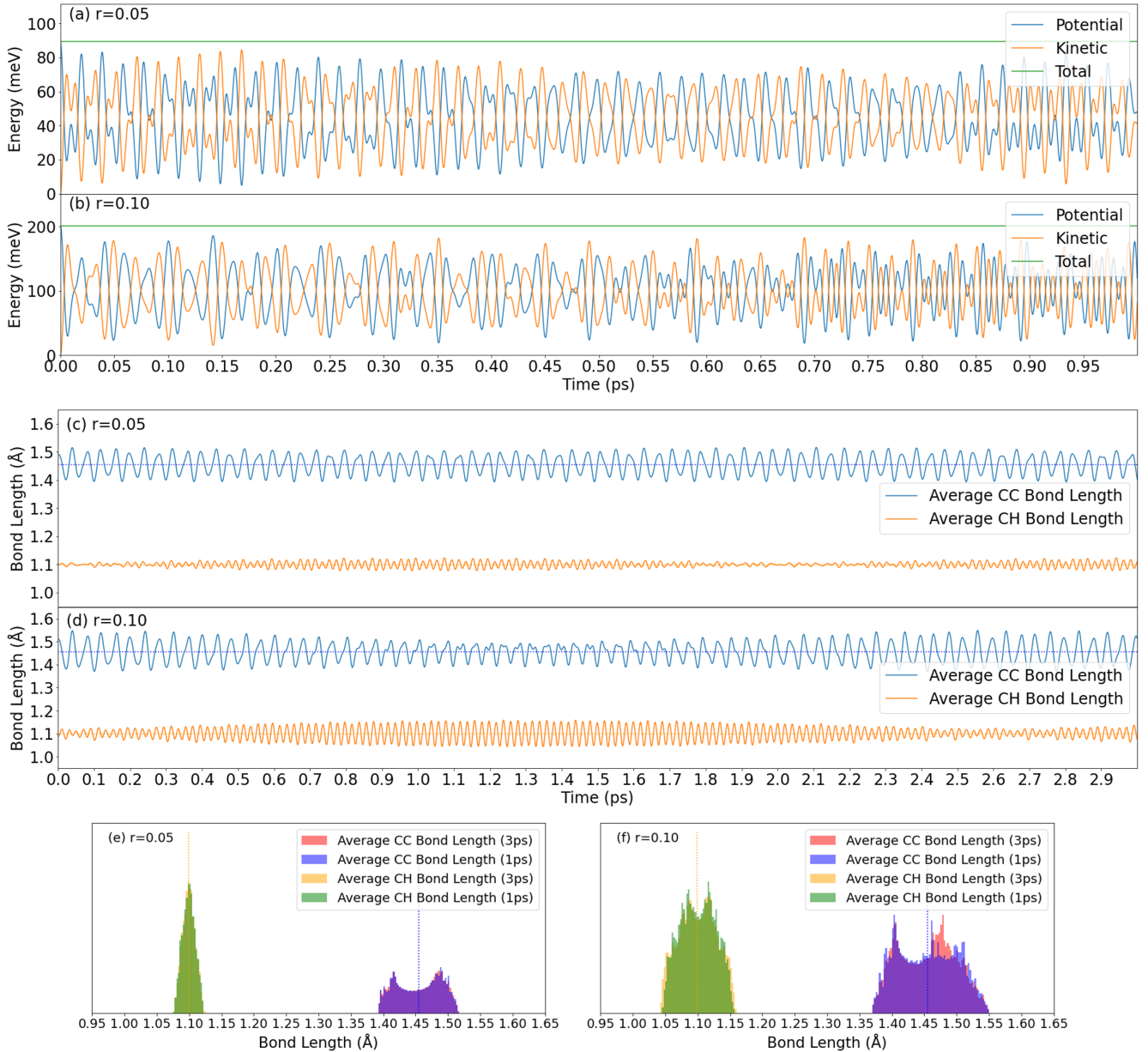
TABLE II: Coefficient and constant values.

We ran two trials for each molecule, one with $r = 0.05$ and another with $r = 0.1$. We examined the first three picoseconds of each trial and produced three graphs: energy vs time, average bond length vs time, and a histograms of the average bond length over that time period.

The energy vs time plot shows how the potential, kinetic, and total energy changes over the first picosecond after initialization. The average bond length vs time plot shows how the average CC and CH bond length changes over the first three picoseconds after initialization. The

histogram of the average bond length displays the probability distribution of the averages of both types of bond lengths for the first picosecond and the first three picoseconds. Both bond length plots show two dotted, horizontal lines that indicate the “expected” bond length (namely $a_0 = 1.455\text{\AA}$ and $b_0 = 1.099\text{\AA}$).

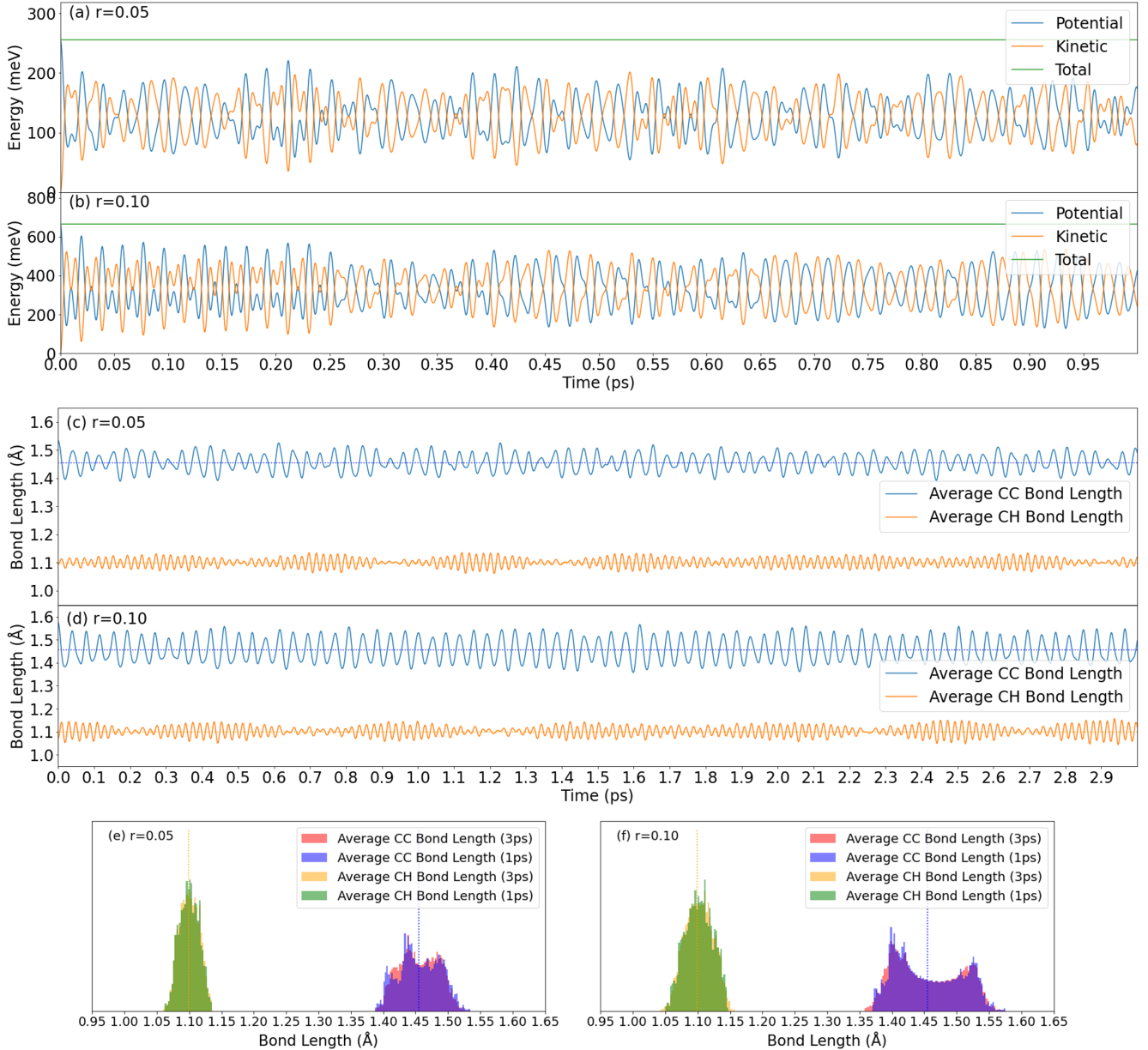
The following pages display these plots. See Appendix E for links to the CSV files of data used to generate these plots, our simulation’s GitHub, and a YouTube video of the simulated molecules.

FIG. 3: Ethane C_2H_6

Ethane (C_2H_6) is the simplest molecule we look at. Consequently, its dynamics are also the most simple. Notably, total energy is constant as it should be in Figure 3a. This is true for all other energy plots as well, and ensures the correctness of our simulation in at least this regard. In Figure 3c, the amplitude of the CC bond fluctuations and the amplitude of the CH bond fluctuations are inversely proportional. When the hydrogen atoms have the majority of kinetic energy, the carbon atoms don't due to the conservation of energy. Similarly, when CC bond

lengths are volatile, CH bond lengths aren't. Furthermore, the period of oscillation is approximately 2ps and the average bond length curves exhibit a pattern analogous to a beat frequency. This is likely because we have two different oscillations corresponding to two different types of springs, the CC bonds and the CH bonds. The distribution of average bond length is virtually identical in both time-periods as evidenced in Figure ??e. The CC bond length tend to have a bimodal distribution, which is unexpected.

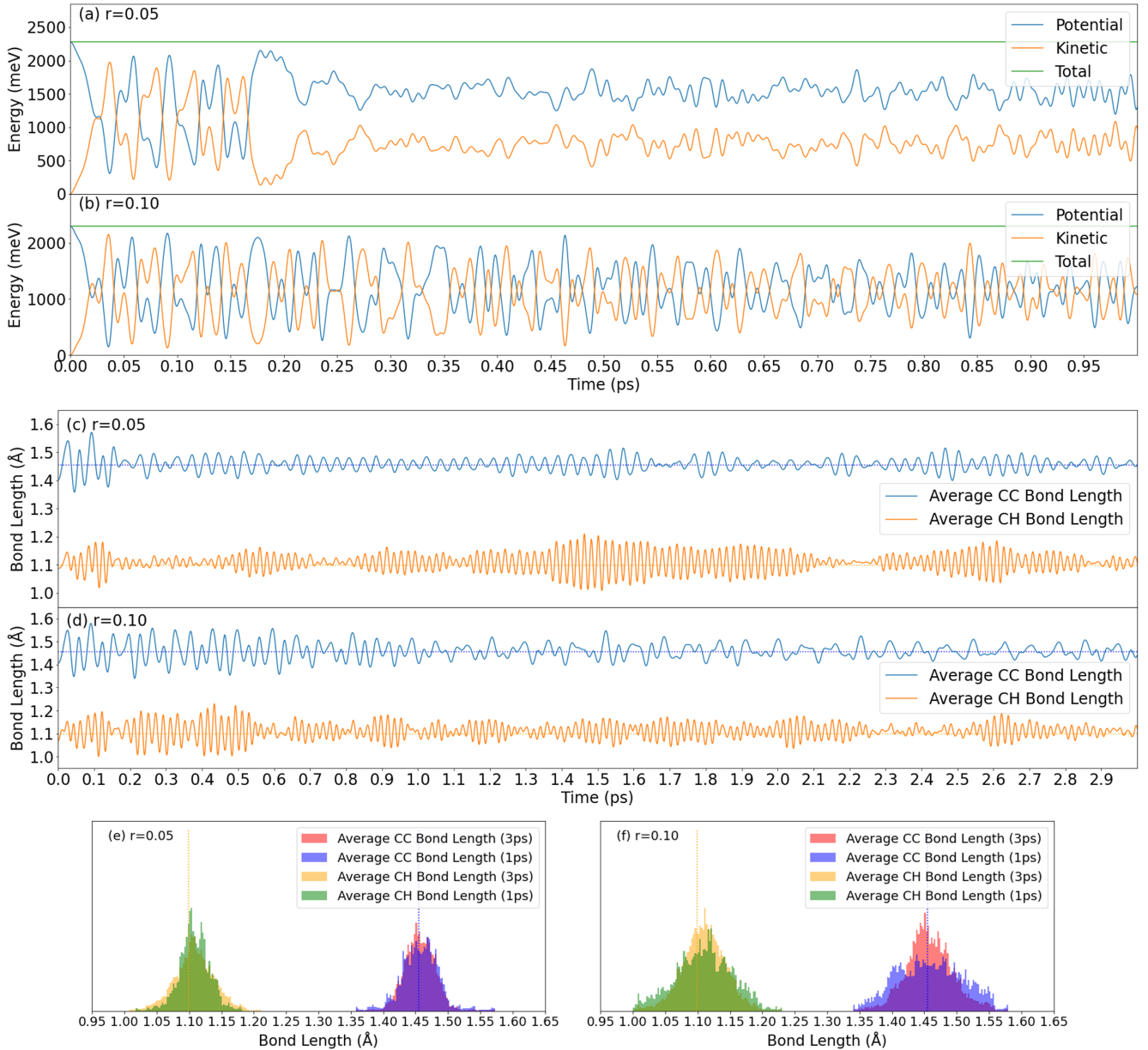
Most of these conclusions seem to hold for $r = 0.1$. However, since we increased the randomization scale, we expect there to be a higher energy than the $r = 0.05$ trial. Likewise, the amplitudes of oscillations for both bond lengths and energy fluctuations increases. We observe that the short-scale frequencies are similar between the two which we would expect but the envelope for the $r = 0.1$ is larger. However, both the CH bond length and CC bond length exhibit bimodal distribution in Figure 3f unlike in Figure 3e. This is probably due to random chance.

FIG. 4: Propane C_3H_8

Propane (C_3H_8), being a larger molecule than ethane, exhibits more complicated oscillations. Examining $r = 0.05$ in Figure 4c, we see that the beat frequency pattern is very obvious. It exhibits a beat frequency behavior, but the dynamics of the envelop is different than that of ethane. This is due to more harmonic oscillators, resulting in more sinusoidal dynamics being summed. We estimate the envelope period to be 0.5ps. The probability distributions do not vary much over time in Figure 4e. The CC bond length distribution is slightly bimodal,

while the CH bond length is Gaussian-like, as expected.

For $r = 0.1$, the CH bond length distribution is Gaussian, but the CC bond length distribution is significantly bimodal. The mean average CH bond length distribution exhibits a lower tail. We might reasonably expect this because hydrogen atoms want to be closer due to its lower electron affinity. Lastly, the bond length plot seems to display steady periodic fluctuations similar to the $r = 0.05$ trial, but with a larger envelope period for the CH bond average.

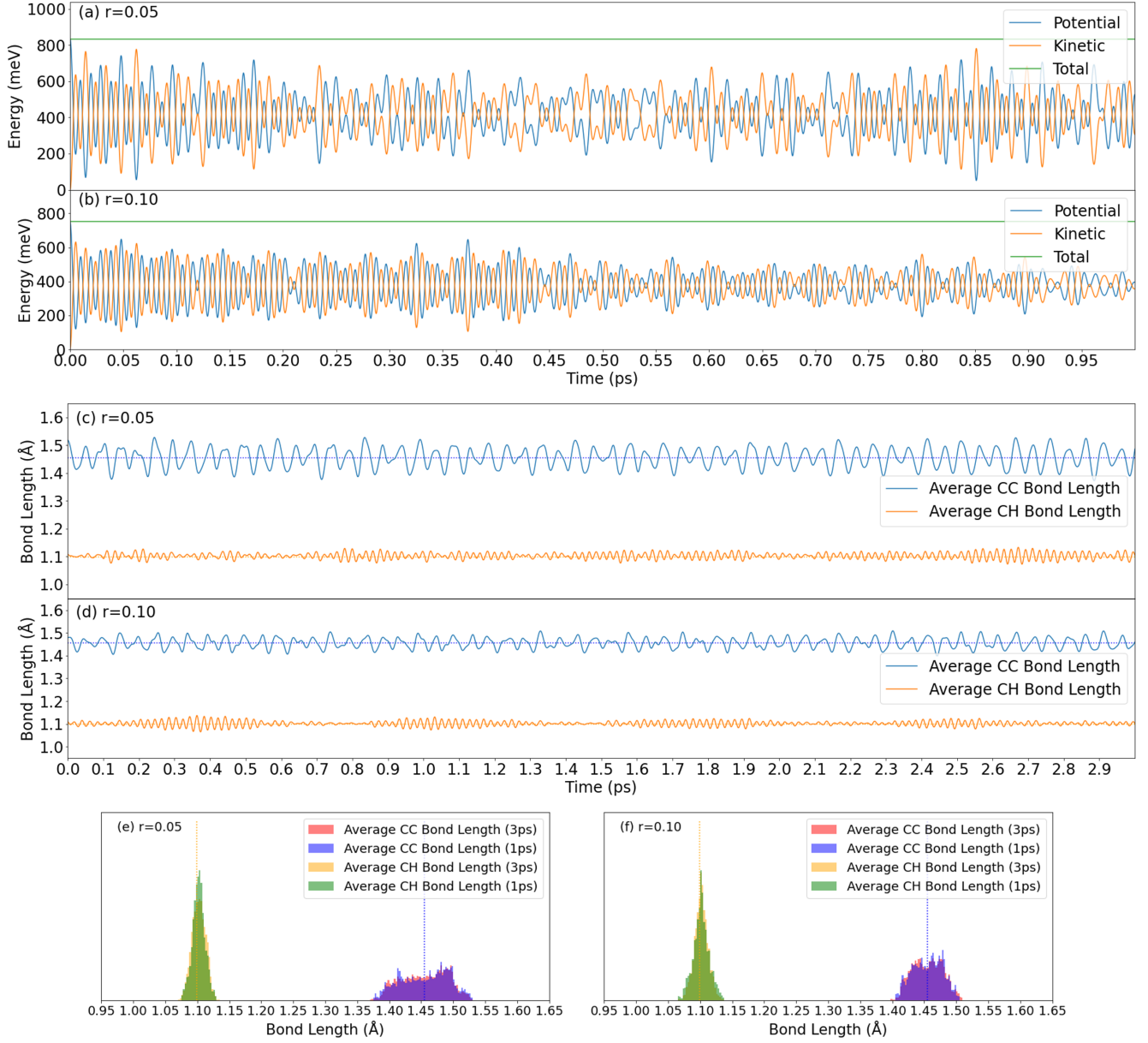
FIG. 5: Benzene C_6H_6

Benzene (C_6H_6) has a ring structure unlike any of the other hydrocarbons we simulated. Furthermore, it has planar symmetry. Unfortunately, due to added complexity, the envelope becomes less periodic in the $r = 0.05$ trials, as evidenced in Figure 5c. In Figure 5a, the energy plot shows rapid vibrations in the first 0.25ps of the simulation but quickly stabilize with potential energy remaining greater than the kinetic energy for the first 3ps. This is further supported by Figure 5e, where the mean of the distribution for CC and CH bonds do not agree with the given parameters distances. Additionally, we

again observe a tail towards lower bond lengths in the CH distribution. Both of these properties are an indication that the molecule's finding an equilibrium where the bond lengths stay further away, this may mean that our coefficients are incorrect for benzene. This would make sense because each carbon atom is super-positioned by a single- and double-bond, unlike other simulated molecules. Notably, in Figure 5e, both bond length histograms are Gaussian-like. The CC bond length distribution changes little over time but the CH bond length distribution attains a slightly larger standard deviation

with time.

The probability distributions are quite different for $r = 0.1$ in Figure 5f. Additionally, in Figure 5b, the energy distribution is considerably different because the kinetic and potential energy settle at the same average. This implies that the atoms have sufficient energies to climb out of the equilibrium positions found in $r = 0.05$. The CC bond length remains Gaussian-like in both time-periods, but the tail for the CH bond length distribution is even more prominent. Furthermore, the width of both bond distributions increases. This change in width is not apparent in simpler molecules such as ethane or propane.

FIG. 6: Isobutane $\text{HC}(\text{CH}_3)_3$

Isobutane ($\text{HC}(\text{CH}_3)_3$) is the largest molecule we considered, and it's almost double the size of ethane with fourteen total atoms. In Figure 6d, the envelope structure is more periodic than benzene. Interestingly, we see that the periodicity of the envelope becomes more defined at later times. This is likely because it takes this larger molecule time to reach its equilibrium dynamics whereas smaller molecules will reach their equilibrium positions much faster due to their size. This is further supported by Figure 6a, where we see the oscillations are more rapid at the early times than for later times. The bond length

probability function, in Figure 6e, shows that the average CC bond length is flat throughout the simulation. The CH bond length retains an approximately Gaussian distribution. We estimate that the envelope period is 0.8ps, though the oscillation doesn't distinctly repeat as well as in the ethane trial.

With $r = 0.1$, most of the properties are identical as the $r = 0.05$ trial. The probability indicates that CC bond length changes very little over time. The average CH bond length's distribution becomes even more narrower over time, unlike before and its mean agrees with our

parameterized $b_0 = 1.099\text{\AA}$, unlike benzene. The period and frequency are approximately 0.7ps.

Interestingly, we see a lower tail on the CH bond length distributions for multiple trials. We are not certain whether Furthermore, the periodicity of the envelopes of the bond length motion becomes shorter as these molecules become larger. This is because we have many more harmonic oscillators operating simultaneously as we add more bonds. As expected, increasing the r value, the widths of distribution in the CH and CC histograms increases. This is because by increasing r adds energy to the system. However, isobutane displays the opposite behavior. We are currently unsure why this is the case. Lastly, we do not expect to see bimodal distributions in the CC or CH distributions. The fact that we see them may indicate an error in our simulation.

There are further improvements to increase the accuracy of our results. Our next steps would be to change the potential to more accurately depict a molecular system. This would be done by using the Morse potential function for the bonded interactions and Lennard-Jones potential function for the nonbonded interactions. All three functions are shown in Figure 7, and the difference is visible for even small deviations around the minima. Since these functions are not harmonic, we are able to go into higher orders of the Taylor expansion. The accuracy of the potential in our current simulation can change drastically as we move away from the equilibrium geometry, as shown in Figure 7. Addressing this issue would increase the scope of possible simulations (and also possibly address the bimodal distributions for CC bonds in a few molecules).

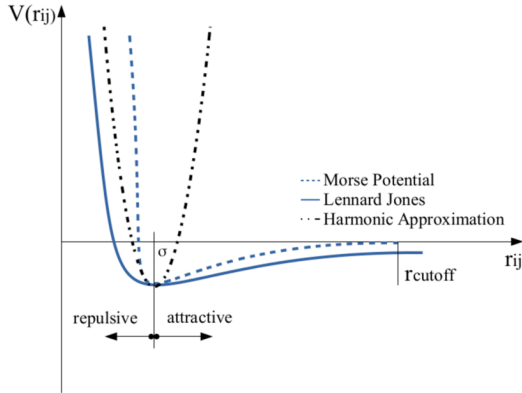


FIG. 7: Morse potential vs Lennard Jones potential vs Harmonic approximation.[7]

Furthermore, currently all our bonds are treated identically while in reality there is a noticeable difference between the properties of single-bonds, double-bonds, and triple bonds. We would need to create and calculate new coefficients to take this into account. Even more generally, it would be preferable to recalibrate our coefficients based off our own experiments rather than relying on Lifson and Warshel's data.

Rest of the V_{bonded} is calculated correctly, but $V_{\text{nonbonded}}$ is just as crucial to calculating V . Unfortunately, $V_{\text{nonbonded}}$ currently isn't even included in our simulation, which is problematic for molecules like benzene which stay in their ground-state largely due to nonbonded interaction. Thus, implementing $V_{\text{nonbonded}}$ would be another possible improvement.

In summary, we developed a consistent force field accurate up to the second order Taylor expansion. For this balls-and-springs methodology, we used the parameters found by Lifson and Warshel[1] as our spring constants. Even with such a simple model, the period of oscillations for all molecules are within one order of magnitude to experimental results found by others[8]. Furthermore, we were able to discern small-scale and large-scale structural changes as is evident by the beat frequency behavior in the CC and CH bond length distributions as well as the changing of their envelope. As noted, there are many improvements that can be made to this simulation in order to investigate configurations further away from the ground state configurations.

Appendix A: Lagrangians

As stated previously, Lagrangians are the result of subtracting potential energy from kinetic, $L = T - V$. Despite seeming nonsensical initially, the equation of motion is yielded by solving the Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = \frac{\partial L}{\partial x}.$$

We will calculate the equation of motion and the normal modes of two objects connected by one spring as an example. We begin with the vector form of the motion (guessing the form of the solutions based on prior knowledge):

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} e^{i\omega t}.$$

Converting to regular equations yields the following:

$$\begin{aligned} -m_1\omega^2 A_1 &= -k(A_1 - A_2), \\ -m_2\omega^2 A_2 &= -k(A_2 - A_1). \end{aligned}$$

Thus, we begin to solve:

$$\begin{aligned} (-m_1\omega^2 + k) A_1 - k A_2 &= 0, \\ -k A_1 + (-m_2\omega^2 + k) A_2 &= 0, \\ \begin{bmatrix} -m_1\omega^2 + k & -k \\ -k & -m_2\omega^2 + k \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \end{aligned}$$

Clearly $\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is one solution, now we proceed to find other solutions. Rephrasing the equation to finding the determinant of the first matrix goes as follows:

$$\begin{aligned} \begin{vmatrix} -m_1\omega^2 + k & -k \\ -k & -m_2\omega^2 + k \end{vmatrix} &= 0, \\ (-m_1\omega^2 + k)(-m_2\omega^2 + k) - (-k)(-k) &= 0 \\ m_1m_2\omega^4 - km_1\omega^2 - km_2\omega^2 + k^2 &= k^2 \\ m_1m_2\omega^4 - km_1\omega^2 - km_2\omega^2 &= 0, \\ \omega^2(m_1m_2\omega^2 - km_1 - km_2) &= 0, \\ \omega &= 0. \end{aligned}$$

This is another solution, and substituting this value of ω

into the matrix to find the following:

$$\begin{aligned} k \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \\ \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} &\propto \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \end{aligned}$$

Other solutions of ω can be found as follows:

$$\begin{aligned} m_1m_2\omega^2 - km_1 - km_2 &= 0, \\ m_1m_2\omega^2 &= km_1 + km_2, \\ \omega^2 &= \frac{km_1 + km_2}{m_1m_2}. \end{aligned}$$

Since ω only appears as ω^2 in the matrices, it is unnecessary to solve for the positive and negative solutions of ω independently. Substitute the above value of ω^2 into the original matrix to find yet another solution:

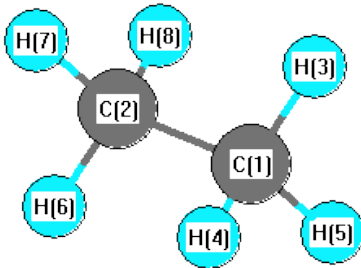
$$\begin{aligned} -k \begin{bmatrix} \frac{m_1}{m_2} & 1 \\ 1 & \frac{m_2}{m_1} \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \\ \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} &\propto \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix}. \end{aligned}$$

Writing all these solutions together yields the following equation of motion:

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = A_s \begin{bmatrix} 1 \\ 1 \end{bmatrix} + A_f \begin{bmatrix} m_2 \\ -m_1 \end{bmatrix} \cos(\omega_f t + \phi_f).$$

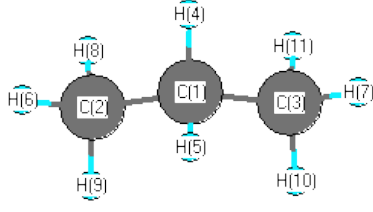
Appendix B: Position Matrices

Listed below are the initial ground state configurations used in our program. The images and values were taken from NIST[9].



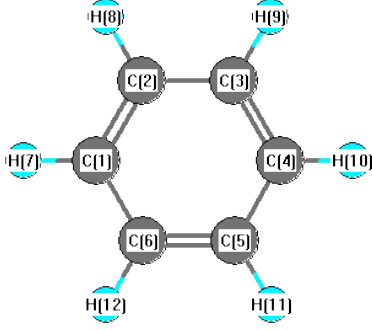
Ethane C_2H_6 matrix =

$$\begin{bmatrix} \text{C1 position} \\ \text{C2 position} \\ \text{H3 position} \\ \text{H4 position} \\ \text{H5 position} \\ \text{H6 position} \\ \text{H7 position} \\ \text{H8 position} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0.7680 \\ 0 & 0 & -0.7680 \\ -1.0192 & 0 & 1.1573 \\ 0.5096 & 0.8826 & 1.1573 \\ 0.5096 & -0.8826 & 1.1573 \\ 1.1092 & 0 & -1.1573 \\ -0.5096 & -0.8826 & -1.1573 \\ -0.5096 & 0.8826 & -1.1573 \end{bmatrix}$$



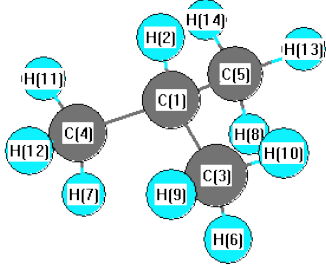
Propane C_3H_8 matrix =

$$\begin{bmatrix} \text{C1 position} \\ \text{C2 position} \\ \text{C3 position} \\ \text{H4 position} \\ \text{H5 position} \\ \text{H6 position} \\ \text{H7 position} \\ \text{H8 position} \\ \text{H9 position} \\ \text{H10 position} \\ \text{H11 position} \end{bmatrix} = \begin{bmatrix} 0 & 0.5863 & 0 \\ -1.2681 & -0.2626 & 0 \\ 1.2681 & -0.2626 & 0 \\ 0 & 1.2449 & 0.8760 \\ -0.0003 & 1.2453 & -0.8760 \\ -2.1576 & 0.3742 & 0 \\ 2.1576 & 0.3743 & 0 \\ -1.3271 & -0.9014 & 0.8800 \\ -1.3271 & -0.9014 & -0.8800 \\ 1.3271 & -0.9014 & 0.8800 \\ 1.3271 & -0.9014 & -0.8800 \end{bmatrix}$$



Benzene C_6H_6 matrix =

$$\begin{bmatrix} \text{C1 position} \\ \text{C2 position} \\ \text{C3 position} \\ \text{C4 position} \\ \text{C5 position} \\ \text{C6 position} \\ \text{H7 position} \\ \text{H8 position} \\ \text{H9 position} \\ \text{H10 position} \\ \text{H11 position} \\ \text{H12 position} \end{bmatrix} = \begin{bmatrix} 0 & 1.3970 & 0 \\ 1.2098 & 0.6985 & 0 \\ 1.2098 & -0.6985 & 0 \\ 0 & -1.3970 & 0 \\ -1.2098 & -0.6985 & 0 \\ -1.2098 & 0.6985 & 0 \\ 0 & 2.481 & 0 \\ 2.1486 & 1.2405 & 0 \\ 2.1486 & -1.2405 & 0 \\ 0 & -2.481 & 0 \\ -2.1486 & -1.2405 & 0 \\ -2.1486 & 1.2405 & 0 \end{bmatrix}$$



Isobutane $HC(CH_3)_3$ matrix =

$$\begin{bmatrix} \text{C1 position} \\ \text{H2 position} \\ \text{C3 position} \\ \text{C4 position} \\ \text{C5 position} \\ \text{H6 position} \\ \text{H7 position} \\ \text{H8 position} \\ \text{H9 position} \\ \text{H10 position} \\ \text{H11 position} \\ \text{H12 position} \\ \text{H13 position} \\ \text{H14 position} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0.3650 \\ 0 & 0 & 1.4730 \\ 0 & 1.4528 & 0.0987 \\ 1.2582 & -0.7264 & -0.0987 \\ -1.2582 & -0.7264 & -0.0987 \\ 0 & 1.4867 & -1.1931 \\ 1.2875 & -0.7433 & -1.1931 \\ -1.2875 & -0.7433 & -1.1931 \\ 0.8941 & 1.9575 & 0.2821 \\ -0.8941 & 1.9575 & 0.2821 \\ 1.2482 & -1.7520 & 0.2821 \\ 2.1422 & -0.2045 & 0.2821 \\ -2.1422 & -0.2045 & 0.2821 \\ -1.2482 & -1.7530 & 0.2821 \end{bmatrix}$$

Appendix C: Ethane Potential Energy Example

As an example, we will manually calculate the potential energy of an ethane (C_2H_6) molecule with the following position matrix:

$$\text{Ethane C}_2\text{H}_6 \text{ matrix} = \begin{bmatrix} \text{C1 position} \\ \text{C2 position} \\ \text{H3 position} \\ \text{H4 position} \\ \text{H5 position} \\ \text{H6 position} \\ \text{H7 position} \\ \text{H8 position} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0.7680 \\ 0 & 0 & -0.7680 \\ -1.0192 & 0 & 1.1573 \\ 0.5096 & 0.8826 & 1.1573 \\ 0.5096 & -0.8826 & 1.1573 \\ 1.1092 & 0 & -1.1573 \\ -0.5096 & -0.8826 & -1.1573 \\ -0.5096 & 0.8826 & -1.1573 \end{bmatrix}.$$

It is important to know that C1 is bonded to C2, H3, H4, and H5; C2 is bonded to C1, H5, H6, H7, and H8; and no other bonds exist. As a reminder, the equation we will be using to approximate the potential energy of hydrocarbon molecules is:

$$\begin{aligned} V(\mathbf{X}) = & \frac{1}{2} \sum_i K_a (a_i - a_0)^2 + \frac{1}{2} \sum_i K_b (b_i - b_0)^2 + \\ & + \frac{1}{2} \sum_i K_\alpha (\alpha_i - \alpha_0)^2 + \frac{1}{2} \sum_i K_\beta (\beta_i - \beta_0)^2 + \frac{1}{2} \sum_i K_\gamma (\gamma_i - \gamma_0)^2 \\ & + \frac{1}{2} \sum_i K_\phi (1 + \cos(3\phi_i)) \\ & + V_{\text{nonbonded}}(\mathbf{X}), \end{aligned}$$

and we will not be calculating $V_{\text{nonbonded}}$ in our simulation. Let's begin by evaluating the first term, the potential energy contributed by CC bonds. There is only one CC bond: between C1 and C2. The length of this CC bond is 1.5360Å thus:

$$\begin{aligned} \frac{1}{2} \sum_i K_a (a_i - a_0)^2 &= \frac{1}{2} \times 573.8 \times (1.5360 \\ &\quad - 1.455)^2 \text{kcal/mol} \\ &= 1.882 \text{kcal/mol}. \end{aligned}$$

Let's proceed similarly for CH bonds. Note that there's 6 such bonds, whose lengths are listed below:

$$\begin{array}{l|l} \overline{\text{C1H3}} & 1.0910\text{\AA} \\ \overline{\text{C1H4}} & 1.0910\text{\AA} \\ \overline{\text{C1H5}} & 1.0910\text{\AA} \\ \overline{\text{C2H6}} & 1.0910\text{\AA} \\ \overline{\text{C2H7}} & 1.0910\text{\AA} \\ \overline{\text{C2H8}} & 1.0910\text{\AA} \end{array}$$

Thus,

$$\begin{aligned} \frac{1}{2} \sum_i K_b (b_i - b_0)^2 &= \frac{1}{2} \times 222.0 \times 6 \times (1.0910 \\ &\quad - 1.099)^2 \text{kcal/mol} \\ &= 0.0426 \text{kcal/mol}. \end{aligned}$$

This calculates the sum of potential energies due to bond lengths. Now let's move onto bond angles, starting with HCH bond angles (since there's no CCC bond angles). Recall that the angle between two vectors u and v can be calculated by $\theta = \arccos\left(\frac{u \cdot v}{|u||v|}\right)$, thus:

$$\begin{array}{l|l} \angle\text{H3C1H4} & 1.8841\text{rad} \\ \angle\text{H3C1H5} & 1.8841\text{rad} \\ \angle\text{H4C1H5} & 1.8888\text{rad} \\ \angle\text{H6C2H7} & 1.8841\text{rad} \\ \angle\text{H6C2H8} & 1.8841\text{rad} \\ \angle\text{H7C2H8} & 1.8888\text{rad} \end{array}$$

Plugging these values into the summation yields the following:

$$\begin{aligned} \frac{1}{2} \sum_i K_\beta (\beta_i - \beta_0)^2 &= \frac{1}{2} \times 76.28 \times (4 \times (1.8841 - 1.911)^2 \\ &\quad + 2 \times (1.8888 - 1.911)^2) \text{kcal/mol} \\ &= 0.1480 \text{kcal/mol}. \end{aligned}$$

Similarly, the list of CCH angles is shown below:

$\angle\text{C1C2H6}$	1.9346rad
$\angle\text{C1C2H7}$	1.9346rad
$\angle\text{C1C2H8}$	1.9346rad
$\angle\text{C2C1H3}$	1.9346rad
$\angle\text{C2C1H4}$	1.9346rad
$\angle\text{C2C1H5}$	1.9346rad

And thus:

$$\begin{aligned}\frac{1}{2} \sum_i K_\gamma (\gamma_i - \gamma_0)^2 &= \frac{1}{2} \times 44.0 \times 6 \times (1.9346 \\ &\quad - 1.911)^2 \text{kcal/mol} \\ &= 0.0735 \text{kcal/mol}.\end{aligned}$$

Finally, we only need to calculate the potential energy contributed by torsional angles and then convert to ap-

propriate units. The torsional angle between vectors u and v , with vector w connecting them, can be calculated as $\phi = \arccos\left(\frac{(u \times w) \cdot (v \times w)}{|u \times w| |v \times w|}\right)$. The 9 torsional angles along with their measurements are listed below:

$\angle\text{H3(C1C2)H6}$	3.1416rad
$\angle\text{H3(C1C2)H7}$	1.0472rad
$\angle\text{H3(C1C2)H8}$	1.0472rad
$\angle\text{H4(C1C2)H6}$	1.0472rad
$\angle\text{H4(C1C2)H7}$	3.1416rad
$\angle\text{H4(C1C2)H8}$	1.0473rad
$\angle\text{H5(C1C2)H6}$	1.0472rad
$\angle\text{H5(C1C2)H7}$	1.0473rad
$\angle\text{H5(C1C2)H8}$	3.1416rad

And now we compute the last summation:

$$\begin{aligned}\frac{1}{2} \sum_i K_\phi (1 + \cos(3\phi_i)) &= \frac{1}{2} \times 2.836 \times 3 \times (1 + \cos(3 \times 3.1416)) + \frac{1}{2} \times 2.836 \times 4 \times (1 + \cos(3 \times 1.0472)) \\ &\quad + \frac{1}{2} \times 2.836 \times 2 \times (1 + \cos(3 \times 1.0473)) \text{kcal/mol} \\ &= \frac{1}{2} \times 2.836 \times (3 \times 0.0000 + 4 \times 0.0000 + 2 \times 0.0000) \text{kcal/mol} \\ &= 0.0000 \text{kcal/mol}.\end{aligned}$$

Finally,

$$\begin{aligned}V(\mathbf{X}) &= \frac{1}{2} \sum_i K_a (a_i - a_0)^2 + \frac{1}{2} \sum_i K_b (b_i - b_0)^2 + \\ &\quad + \frac{1}{2} \sum_i K_\alpha (\alpha_i - \alpha_0)^2 + \frac{1}{2} \sum_i K_\beta (\beta_i - \beta_0)^2 + \frac{1}{2} \sum_i K_\gamma (\gamma_i - \gamma_0)^2 \\ &\quad + \frac{1}{2} \sum_i K_\phi (1 + \cos(3\phi_i)) \\ &= 1.882 + 0.0426 + 0 + 0.1480 + 0.0735 + 0.0000 \text{kcal/mol} \\ &= 2.1461 \text{kcal/mol} \\ &= 14.9100 \text{zJ}.\end{aligned}$$

Appendix D: JAX

JAX has four incredible program transformations for writing numerical code such as this simulation:

1. jit, for speeding up code
2. grad, for taking derivatives
3. vmap, for automatic vectorization or batching

4. fori_loop, for speeding up loop operations

Let's discuss these in greater detail. JAX runs primarily on the GPU if one is available; if there is none, then JAX will run on the CPU. Normally, JAX dispatches kernels to the GPU/CPU one operation at a time. However, jit compiles multiple operations together using XLA, making code up to 5 times faster.

Traditionally, derivatives of functions would be cal-

culated by hand and then hard-coded into the program. Along with being inefficient, this method would also be error-prone and difficult to update for newer, perhaps more accurate, formulas. JAX’s grad function remedies this by calculating the derivative of any numerical function programmed with JAX in an instant with incredible accuracy.

Due to the excessive amount of number-crunching needed to run a successful simulation, it makes sense to store information in arrays. However, JAX’s arrays are just matrices with a new name. This allows us to do matrix operations (such as updating $\mathbf{X}$ to $\mathbf{X} + \dot{\mathbf{X}}\Delta t + \frac{\ddot{\mathbf{X}}(\Delta t)^2}{2}$), but, even more importantly, this allows us to use vmap. The vmap transformation has similar semantics of mapping a function along array axes, but instead of keeping the loop outside, vmap pushes the loop into a function’s primitive operations for better performance - up to a factor of 30.

Lastly, in any simulation, it is necessary to call the update function repeatedly. Without `fori_loop`, this was done outside of the molecule class. Switching back and forth is time-intensive, `fori_loop` essentially removes the middleman. Specifically, the outside loop calls the `fori_loop` method (which rapidly calls update 100,000

times and records the data every 100 updates into a temporary array), then copies data from the temporary array to a public array, then empties the temporary array, then the process repeats. Obviously this is better than the outside loop calling update 100 times, then recording, then repeating. `fori_loops` improved speed by a factor of 10.

Before implementing the optimizations listed above, it took about 1 hour to run 100,000 update calls. With these optimizations in place, however, the simulation could run 100,000,000 update calls in 40 minutes. Thus, armed with JAX, creating powerful physical simulations on Python has become very practical.

Appendix E: CSV Files, GitHub, and YouTube

1. CSV Files: https://drive.google.com/file/d/1DCI9PWngpDC0wGWAZNWLLJBHga-x_eFh*
2. GitHub: <https://github.com/f4t4nt/MolSim>
3. YouTube: <https://youtu.be/iCUkThONkhc>

*7-zip or equivalent required to open files due to large size.

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